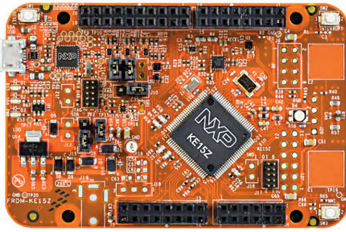


consumes extremely low power despite having a few components that are in general power-hungry.

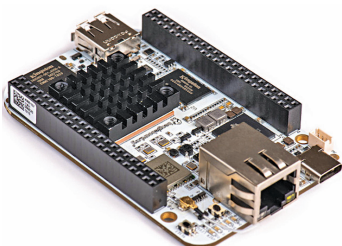
5. FRDM-KE15Z. The Freedom Development board range from NXP is appropriate for embedded hardware and IoT development



(Source: NXP Semiconductors)

and is packed with NXP sensors. Each board has different features, and hence an engineer can select a board according to a particular application. The FRDM-KE15Z board contains an RGB LED, a 3-axis gyroscope, a temperature sensor, and two capacitive touchpads. Also, the NXP community is a good platform for beginners to post their queries and get them solved by experienced developers.

6. BBONE-AI. In several applications, IoT, ML, and AI go hand in hand. The BeagleBone AI is particularly suited for embedded artificial

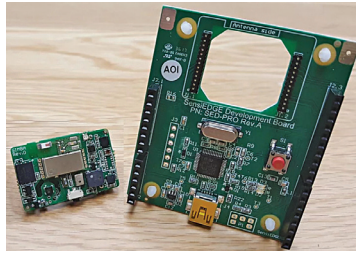


(Source: BeagleBoard Org)

intelligence and has become popular in the BeagleBoard community. It allows users to implement neural networks and TensorFlow models. Moreover, it has programmable real-time units (PRUs) that can be used for time-sensitive applications while the intense computing tasks are done on its ARM processor.

7. SIMBA-DKL. It has all the necessary features needed for IoT

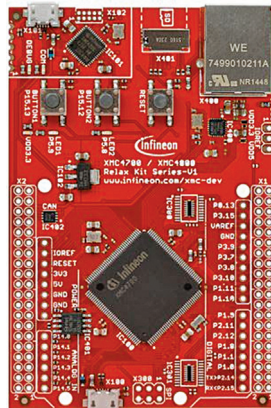
prototyping and embedded product development including light sensor, pressure sensor, accelerometer,



(Source: SensiEdge)

gyroscope, and BLE (Bluetooth Low Energy) connectivity. Since it is compatible with Arduino IDE, you can leverage your knowledge in the Arduino framework to make innovative projects.

8. KITXMC47RELAXV1TOB01. Commonly known as the XMC4700 Relax Kit, this board can be used as an Industrial IoT (IIoT) node or as a

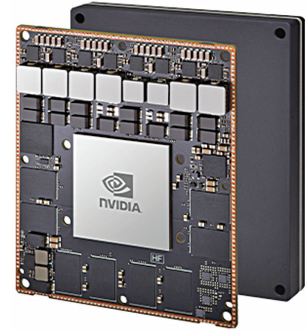


(Source: Infineon Technologies)

gateway in commercial applications. It has an onboard micro SD card slot that not just expands the memory but can also be used to ensure that data is stored locally on the device, thus preventing data loss. Infineon has also created several user groups on its community website where developers can participate in discussions and post queries regarding this board, if required.

9. Nvidia AGX Xavier Industrial. It promises fast AI development and AI inferencing capabilities on edge devices. It can be useful for applications involving edge

computing, which is significant in modern AI devices these days. It has an operating lifetime of ten years

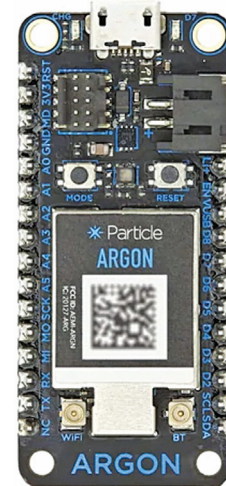


(Source: Nvidia)

and works on extremely low power. Smart cities, logistics, and autonomous robots are desirable application areas for this board. Nvidia also offers AI certifications to those who submit their Jetson based projects to the Jetson developer community.

10. Particle Argon. This board is known for its usage in Wi-Fi applications. It allows the crea-

tion of mesh networks. This means that even if one device is connected to the internet, all devices in the network are connected to the internet. You can even power it using a LiPo (lithium-ion polymer) battery. Programming and trouble-



(Source: Particle IO)

shooting the board is extremely easy due to the user-friendly Particle Web IDE and the Particle community.

Detailed comparison of these boards is given in Table 1. **EFY**

The author, Aaryaa Padhyegurjar, is an Industry 4.0 enthusiast with a keen interest in innovation and research.